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EDITED BY
Mohan Mondal,
ICAR-National Dairy Research Institute,
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REVIEWED BY
Javier Piloni Martini,
Autonomous University of Hidalgo State,
Mexico
Yu Wang,
Northwest A&F University, China

*CORRESPONDENCE
Iurii Stafeev
yuristafeev@gmail.com

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Adaptive glucose homeostasis in the desert rodent *Acomys cahirinus*: high glucose tolerance, liver metabolic shifts, and visceral adiposity

Maria Boldyreva[1,2], Rita Mohammad[1,3], Margarita Agareva[1,4], Svetlana Michurina[1], Natalia Alekseeva[5], Elizaveta Ratner[1], Yelena Parfyonova[1,4] and Iurii Stafeev[1]*

[1]Chazov National Medical Research Centre for Cardiology, Moscow, Russia, [2]National Research University Higher School of Economics, Moscow, Russia, [3]Moscow Institute of Physics and Technology, Dolgoprudny, Russia, [4]Faculty of Basic Medicine, Lomonosov Moscow State University, Moscow, Russia, [5]Faculty of Biology, Lomonosov Moscow State University, Moscow, Russia

Background: Mammals from arid ecosystems face distinct physiological challenges: water scarcity and unpredictable food availability require the integration of energy storage, maintenance of water balance, and flexible switching between nutrients. Our previous study demonstrated lower fasting blood glucose levels in the desert rodent *Acomys cahirinus* (African spiny mouse), which also possesses a unique regenerative ability. The aim of this study was to evaluate selected aspects of systemic glucose tolerance and its mechanisms in this species.
Methods: Healthy *Acomys cahirinus* and *Mus musculus* mice, used as a classic laboratory rodent control, were maintained under the same conditions in an SPF vivarium. Systemic metabolism was evaluated using insulin and glucose tolerance tests. Pancreas physiology was analyzed by immunohistochemistry. Liver state was assessed using tolerance tests with gluconeogenic substrates and histological analysis. Glucose uptake and its metabolic fate were analyzed using radioisotope tracing.
Results: We have shown that Acomys cahirinus has higher glucose tolerance. The pancreas of *Acomys cahirinus* is characterized by hyperplastic remodeling with a predominance of insulin-producing cells. *Acomys cahirinus* shows lower gluconeogenic activity based on pyruvate and glycerol, but it is enhanced on alanine. *Acomys cahirinu*s takes up glucose predominantly in visceral fat depots, with consequent lipogenesis. In the fed state, *Acomys cahirinus* prefers to store glycogen in the liver, whereas after starvation it preferentially stores lipids.
Conclusions: *Acomys cahirinus* is a desert rodent with adaptations to arid environments that are manifested as enhanced glucose tolerance and altered pancreatic, hepatic, and adipose tissue physiology. Some of these mechanisms can be used as a potential strategy for the treatment of human metabolic diseases.

KEYWORDS
Acomys cahirinus, adipose tissue, glucagon, glucose tolerance, insulin, liver physiology, pancreas, spiny mice

1 Introduction

Mammals inhabiting arid ecosystems face distinct physiological challenges: water scarcity and unpredictable food availability require the integration of energy storage, maintenance of water balance, and flexible switching between nutrients (Schmidt-Nielsen, 1997; Willmer et al., 2005; Fuller et al., 2014). Under such conditions, the regulation of carbohydrate metabolism may have evolved along adaptive trajectories distinct from those observed in the "classical" laboratory mouse model, in which glycemic maintenance is viewed primarily through the lens of skeletal muscle insulin sensitivity and the risk of metabolic dysfunction (Nilsson et al., 2006). Thus, the comparative physiology of arid-adapted species offers an opportunity to identify alternative and potentially adaptive solutions to the challenge of glucose homeostasis (Schwimmer and Haim, 2009).

The African spiny mouse (*Acomys cahirinus*) is a desert rodent capable of complete regeneration of skin and skeletal muscle (Seifert et al., 2012), in addition to functional recovery following injury to the heart, kidney, and spinal cord (Allen and Seifert, 2025). Regeneration is an energetically expensive process and likely requires a specialized organization of carbohydrate metabolism. Previous studies have demonstrated that *Acomys* sp. exhibit consistently low fasting glycemia (3.8–7.5 mM, depending on fasting duration) and greater glucose tolerance compared with conventional laboratory models, including C57BL/6 mice (Gutzeit et al., 1974; Boldyreva et al., 2025). These observations suggest that the fundamental organization of metabolism in spiny mice may differ substantially from that of standard laboratory rodents. According to laboratory observations, Acomys cahirinus can develop obesity under laboratory conditions, whereas in the wild Acomys cahirinus has a fiber-rich and protein-rich diet with nutrition from plants and insects (Crews et al., 2024). Systemic manifestations, such as low fasting glucose and rapid glucose clearance, could arise through various physiological mechanisms.

First, fasting glycemia is determined by the balance between endogenous glucose production by the liver through gluconeogenesis or glycogenolysis and exogenous glucose uptake. Second, glucose dynamics depend on the capacity and endocrine architecture of the pancreas, the pattern of insulin response, and the activity of glucose fluxes among tissues (e.g., between muscle and distinct adipose depots) (DeFronzo et al., 1992). A thorough understanding of the contributions of these regulatory levels is critical for correctly interpreting "low glycemia" as an adaptation rather than as a manifestation of a pathological state.

The liver is the main orchestrator in maintaining fasting glycemia, governing the rate of glucose entry into the circulation and the selection of substrates for gluconeogenesis (lactate/pyruvate, glycerol, and glucogenic amino acids) (Roden and Shulman, 2019). Simultaneously, the pancreas determines the efficiency with which hepatic glucose production is suppressed and the rate of glucose utilization by peripheral tissues following nutrient intake (Petersen and Shulman, 2018). Finally, peripheral tissues not only "consume" glucose but also dictate its fate: oxidation in muscle or storage as lipids in adipose tissue (Cinti, 2022).

In contrast to the traditional paradigm in which low glycemia is attributed to enhanced insulin sensitivity, we hypothesize that the low fasting glycemia characteristic of normoglycemic spiny mice arises not from globally increased insulin sensitivity but rather from a coordinated reorganization of the "liver–pancreas–peripheral tissues" axis. This reorganization includes (i) a distinct organization of the pancreatic endocrine apparatus, (ii) specific tuning of hepatic glucose production and gluconeogenic substrate preference, and (iii) a shift in glucose utilization away from muscle and toward adipose tissue, which may enable more efficient mobilization of energy reserves under conditions of unpredictable food availability.

The goal of our study was to identify species-specific features of glucose metabolism organization in *Acomys cahirinus* through direct comparison with the *Mus musculus* C57BL/6 strain. Understanding these physiological mechanisms may illuminate alternative strategies for carbohydrate regulation and may have implications for the development of therapeutic approaches for metabolic diseases.

2 Materials and methods

2.1 Animal husbandry

Healthy male *Mus musculus* C57BL/6J mice (weight 20–22 g) were obtained from the "Andreevka" animal husbandry facility (Russia) and acclimated in the vivarium of the Chazov National Medical Research Center of Cardiology. Healthy male *Acomys cahirinus* (weight 33–34 g) were obtained from our in-house breeding colony. Animals were housed in cages, 2–3 animals per cage. Male animals were used in this study to minimize metabolic variability, as recommended for metabolic research, due to the absence of the estrous cycle, which can influence metabolic parameters in females. Animals were randomly assigned to experimental groups (n = 10 per group). All experiments were performed using these animal populations. Baseline characteristics of experimental animals are presented in Table 1.

All animals were housed in ventilated cages at 27 ± 2 °C under a 12h/12h light/dark cycle in standard pathogen-free conditions, with sterile chow and water provided *ad libitum*. The diet for C57BL/6J mice consisted of a standard diet for laboratory rodents (#PK120, Gatchina Animal Complete Feeds Factory, Russia). The diet for *Acomys cahirinus* consisted of the same standard diet for laboratory rodents supplemented with dried *Hermetia illucens* imagoes (three per week per animal). Animal experiments were performed in accordance with the internal "Rules for conducting work using experimental animals" and the ARRIVE guidelines and were approved by the Ethical Board of the Chazov National Medical Research Center of Cardiology (permit # 3/24, 03.04.2024). All obtained results were included in the analysis without exclusions.

Abbreviations: ARRIVE, Animal Research: Reporting of *In Vivo* Experiments; ATT, alanine tolerance test; AUC, area under the curve; DMEM, Dulbecco's Modified Eagle Medium; GTT, glucose tolerance test; GlyTT, glycerol tolerance test; ITT, insulin tolerance test; PTT, pyruvate tolerance test; PBS, phosphate-buffered saline; RIPA, radioimmunoprecipitation assay; SEM, standard error of the mean; SFV, phosphosulfovanilline reagent.

TABLE 1 Baseline phenotypic characteristics of experimental animals.

Parameter	*Mus musculus*	*Acomys cahirinus*
N, male animals	10	10
Age, weeks	8–9	8–9
Body weight, g (mean ± SEM)	21,0 ± 0,5	33,5 ± 0,4
Fasting blood glucose, mM (mean ± SEM)	4,24 ± 0,18	3,23 ± 0,11

The body weight of animals was measured using a standard laboratory weigh scale (Mettler-Toledo, Switzerland). Blood glucose levels were measured using a Contour TS glucometer (Ascensia Diabetes Care, Basel, Switzerland).

2.2 Glucose tolerance and insulin sensitivity assays

To evaluate glucose tolerance, we performed a glucose tolerance test (GTT). Animals were fasted for 18 h and weighed before the measurements. Blood samples were collected from the tail vein by cutting the tip of the tail with sterile scissors, and blood glucose levels were assessed using a Contour TS glucometer. For the GTT, 2 g/kg of glucose was injected intraperitoneally in the form of a 20% sterile glucose solution. Blood glucose levels were measured before glucose injection (0 min) and at 15, 30, 45, 60, 90, and 120 min after glucose administration. For the insulin sensitivity evaluation, we performed an insulin tolerance test (ITT). Animals were fasted for 6 h and weighed before the measurements. Blood samples were collected from the tail vein by cutting the tip of the tail with sterile scissors, and blood glucose levels were assessed using a Contour TS glucometer. For the ITT, 0.25 IU/kg of human insulin (Humulin, Lilly France GmbH, France) was injected intraperitoneally. Blood glucose levels were measured before insulin injection (0 min) and at 15, 30, 45, 60, 90, and 120 min after insulin administration. At the end of both GTT and ITT, we created a curve and calculated the parameter area under the curve (AUC) according to Gagnon's method (Gagnon and Peterson, 1998).

2.3 Pancreas histology analysis

We surgically isolated the pancreas from animals and fixed it in 10% paraformaldehyde with further mounting in a paraffin block. Tissue samples were sliced into 5 uM sections using a microtome. Slides were deparaffinized and hydrated up to distilled water. Tissue sections were fixed in 4% paraformaldehyde solution, washed, and incubated in blocking solution (2% bovine serum albumin, 10% of the secondary antibody's donor serum, and PBS). Slides were incubated overnight with primary antibodies (anti-insulin, #I2018, Sigma-Aldrich, USA; anti-glucagon, #2760, Cell Signaling, USA). After incubation with primary antibodies, the sections were stained with Alexa Fluor 488-conjugated secondary antibody (#A21206, Thermo Scientific, USA) or with Alexa Fluor 594-conjugated secondary antibody (#A21203, Thermo Scientific, USA). All slides were counterstained with DAPI (Sigma-Aldrich, USA). Staining was visualized on a Leica Stellaris 5 confocal microscope (Leica Microsystems, Germany), and section were analyzed using ImageJ freeware.

2.4 *Ex vivo* hepatic glucose production assay

One of the glucose production processes in the whole body is hepatic gluconeogenesis. Given the differences in blood glucose levels between *Mus musculus* and *Acomys cahirinus*, we analyzed glucose production by hepatic explants under ex vivo conditions, as previously described (Rosebrough and Steele, 1987). Tissue samples (liver explants) were collected from animals under avertin anesthesia (2.5% solution, i.p.) under sterile surgical conditions, washed three times with warm PBS, put into glucose-free DMEM or glucose-free DMEM supplemented with 1 mM sodium pyruvate (all media were supplemented with PenStrep and 10% FBS), and incubated for 24 h. After incubation, conditioned media were collected and glucose concentrations were analyzed enzymatically using a hexokinase-based D-Glucose HK Assay Kit (K-GLUHK-110A, Megazyme, Ireland) according to the manufacturer's instructions. Liver explants were weighed using an analytical weight scale, and the values were used for glucose concentration normalization.

2.5 *In vivo* gluconeogenic substrate assays

To evaluate gluconeogenesis at the systemic level, we performed metabolic tests using bolus injections of gluconeogenic substrates, which can be obtained from carbohydrates (pyruvate), lipids (glycerol), and proteins (alanine) (Benedé-Ubieto et al., 2020). All tests were performed after overnight fasting. Pyruvate was injected in the form of a 20% solution at a dose of 2 g/kg, and the same for glycerol - 20% solution, 2 g/kg, whereas alanine was administered as a 15% solution. Pyruvate and glycerol were purchased from Sigma-Aldrich, USA, and alanine was purchased from PanEco, Russia. Blood samples were collected from the tail vein by cutting the tip of the tail with sterile scissors, and blood glucose levels were assessed using a Contour TS glucometer. Blood glucose levels were measured before bolus injection (0 min) and at 15, 30, 45, 60, 90, 120, and 150 min after bolus injection. At the end of the tests, we created a curve and calculated the parameter area under the curve (AUC) according to Gagnon's method (Petersen and Shulman, 2018).

2.6 Glucose uptake

Overnight-fasted animals (n = 5) were injected with a mix of 2 g/kg glucose (Sigma-Aldrich, USA) and 1 µCi radioactive non-metabolizing analog [3H]-2-deoxyglucose (#ART0103B, American Radiolabeled Chemicals, USA) per animal. After 2 h of injection, subcutaneous and epididymal adipose tissue samples were collected, immediately frozen in liquid nitrogen, and stored at –70 °C. Before further sample processing, frozen samples were weighed using an analytical weight scale for further normalization. Tissue samples were homogenized on ice in RIPA buffer (50 mM Tris-HCl, pH 8.0; 150 mM NaCl; 1% Triton X-100; 0.5% sodium deoxycholate; 0.1% sodium lauryl sulfate) at a ratio of 3 µL buffer per mg of tissue using a drill-assisted pestle homogenizer (Eppendorf, Germany). Samples were centrifuged at 4 °C and 5000g for 5 min, and the supernatant was used for further analysis. Supernatant samples were mixed with scintillation liquid (Unisolve100, #95691; Koch Light Ltd., Germany). Radioactivity was measured as counts per minute (CPM) using a TRI-CARB 4910TR scintillation counter (PerkinElmer, USA).

2.7 Lactate concentration biochemical assay

For the control of glycolysis activity, we analyzed lactate production in the general bloodstream and in subcutaneous and epididymal fat. Overnight-fasted animals (n = 5) were injected with 2 g/kg glucose (Sigma-Aldrich, USA) per animal. After 2h of injection, subcutaneous and epididymal adipose tissue samples were collected, immediately frozen in liquid nitrogen, and stored at −70°C. In plasma samples, lactate concentrations were measured directly. Tissue samples were homogenized on ice in a PBS buffer at a ratio of 3 µL buffer per mg of tissue using a drill-assisted pestle homogenizer. Samples were centrifuged at 4 °C and 5000g for 5 min, and the supernatant was used for further analysis. Lactate concentrations in samples were assessed enzymatically using the L-Lactic Acid (L-Lactate) Assay Kit (K-LATE, Megazyme, Ireland) according to the manufacturer's instructions.

2.8 [^{14}C]-metabolic tracing

Overnight-fasted animals (n = 5) were injected with a mix of 40 mg [^{12}C]-glucose (Sigma-Aldrich, USA) and 2 µCi radioactive [^{14}C]-labeled glucose (#ARC0122G; American Radiolabeled Chemicals, USA) per animal. After 24 h of injection, subcutaneous and epididymal adipose tissue samples were collected, immediately frozen in liquid nitrogen, and stored at −70°C. For metabolite extraction, tissue samples were homogenized on ice in RIPA buffer (50 mM Tris-HCl, pH 8.0; 150 mM NaCl; 1% Triton X-100; 0.5% sodium deoxycholate; 0.1% sodium lauryl sulfate) at a ratio of 3 µL buffer per mg of tissue using a drill-assisted pestle homogenizer. Metabolite fractionation was performed as described previously (Michurina et al., 2024). Briefly, 500 µL chloroform/methanol (2:1) mixture was added to homogenates, followed by incubation on a shaker for 10 min. Samples were centrifuged at 5000 g for 5 min to separate the two phases. The lower phase contained the total lipid fraction, whereas the upper water soluble phase contained the water-soluble metabolite fraction, which contains water-soluble metabolites from cytoplasm, mitochondria, and other cellular compartments. The extraction of lipids and water soluble metabolites from lysates was performed twice. Each phase was mixed separately with scintillation liquid. Radioactivity was measured as counts per minute (CPM) using a TRI-CARB 4910TR scintillation counter. Radioactivity data were normalized on tissue sample weight.

2.9 Glycogen histological staining

We isolated tissue samples from the liver; the samples were fixed in 10% paraformaldehyde and mounted in a paraffin block. Tissue samples were sliced into 5 µm sections using a microtome. Slices were deparaffinized and hydrated up to distilled water. Staining was performed using the Glycogen staining kit (HK-PS-AQC5, BioVitrum, Russia). Slices were transferred to periodic acid solution for 10 min, followed by double washing in distilled water. Subsequently, slices were transferred to Schiff's reagent for 20 min, rapidly washed with distilled water, and immersed twice in sulfuric acid solution for 2 min each. Slices were counterstained with Harris hematoxylin. Finally, slices were then dehydrated in graded ethanol solutions (70%, 96%, and 100%), followed by xylene (10 min each). Sections were mounted in Cytoseal-60 (Richard-Allen Scientific, USA) under coverslips. Stained slides were scanned using a Leica ScanScope CS microscope (Leica Microsystems, Germany) and analyzed with Aperio ImageScope software.

2.10 Oil Red O histological staining

After skin dissection, the liver samples were harvested and frozen in TissueTek medium (Sakura Finetek, Netherlands). Tissue sections (7 µm thick) were prepared on glass slides and stored at −80 °C. For each animal, four different sections were analyzed, with a total of 10 animals per group. Tissue sections were fixed in 4% paraformaldehyde and washed with distilled water. Staining by Oil Red O was performed as previously described (Mehlem et al., 2013). Briefly, tissue sections were placed on glass slides and were incubated in Oil Red O working solution for 5 min. Samples were then washed three times in distilled water for 30 min and incubated in 70% ethanol, 96% ethanol, and 100% ethanol, followed by xylene (10 min). Slides were mounted in Cytoseal-60 under coverslips. Staining was visualized on a Leica ScanScope CS scanning microscope and analyzed using Aperio ImageScope software. We analyzed five fields of view for each section. Stained area analysis was performed using ImageJ (NIH, USA). For each TIFF image (8-bit, red channel), the threshold was manually adjusted to ensure that all visible Oil Red O-positive structures (lipid droplets stained orange-red) were accurately identified, whilethe background was excluded. A single fixed threshold, selected based on a representative section from the control group, was used for all comparison groups. The final results are presented using a conservative manual threshold that minimizes artifacts (air bubbles or tissue irregularities). All thresholding steps were performed in a blinded manner.

2.11 Total lipids biochemical assay

For total lipid analysis, liver samples were homogenized on ice in a chloroform/methanol mix at a ratio of 10 µL mixture per mg of tissue using a drill-assisted pestle homogenizer. Samples were centrifuged at 4 °C and 10000g for 10 min. Subsequently, 100 µL of supernatant was transferred to a separate tube and dried under a nitrogen stream using a Bunsen valve. We then used a sulfovanillin-based method for the lipid concentration assessment, as previously described (Bailey et al., 2022). Briefly, 10 µL of concentrated H_2SO_4 was added, followed by incubation at 90 °C for 2 min and cooling on ice to room temperature. Then, we added SPV reagent (1.2 mg/mL vanillin water solution supplemented by concentrated H_3PO_4) up to a volume of 1 mL, and samples were incubated with gentle mixing for 15 min. Optical density was analyzed at 490 nm using an INNO-S plate reader (LTek, South Korea).

2.12 Statistical assay

Data are shown as mean ± standard error of the mean (SEM) for each group. Software GraphPad Prism 8.0 was used for the statistical analysis and graph imaging. Significance of paired differences was calculated using the Mann-Whitney rank sum U-test, with a statistical significance threshold of $p<0.05$.

3 Results

3.1 *Acomys cahirinus* has higher glucose tolerance than *Mus musculus*

Our previous study focused on the analysis of adipose-derived stem cells from *Acomys cahirinus* and selected physiological features of its adipose tissue. We demonstrated that fasting blood glucose levels in *Acomys cahirinus* were lower than those in *Mus musculus* (Boldyreva et al., 2025). In the present study, we extended these observations by performing systemic metabolic tests to evaluate glucose tolerance and insulin sensitivity.

The GTT reflects animal tolerance to glucose loading and demonstrates glucose uptake rate from the general bloodstream. We demonstrated that *Acomys cahirinus* exhibited a higher rate of glucose clearance from the bloodstream, with significant differences in blood glucose levels observed at 15 and 30 min of the GTT (Figure 1A). In addition, the AUC of the GTT was significantly lower in *Acomys cahirinus* than in *Mus musculus* (Figure 1B). Insulin sensitivity revealed the opposite pattern. *Acomys cahirinus* demonstrated lower insulin sensitivity on the ITT curve in early response (15 and 30 min), whereas on the long period curve, insulin sensitivity was similar for both animals (Figure 1C). The AUCs of the ITT of *Acomys cahirinus* and *Mus musculus* had no statistically significant difference (Figure 1D). In summary, the obtained results indicate that *Acomys cahirinus* exhibits higher glucose tolerance and glucose clearance rate than *Mus musculus*. The physiological mechanisms underlying this enhanced glucose clearance rate in *Acomys cahirinus* require further investigation.

3.2 The pancreas of *Acomys cahirinus* has shifted morphology for high production of insulin and glucagon

Potential mechanisms that can enhance the glucose clearance rate may involve altered pancreas physiology. We surgically isolated the pancreas from *Acomys cahirinus* and *Mus musculus* and analyzed pancreatic morphology using immunohistochemical staining.

Immunohistochemical staining demonstrated several interesting aspects of *Acomys cahirinus* pancreatic morphology. First, *Acomys cahirinus* exhibited enlarged Langerhans islets and islet density, which can be evidence of enhanced hormone-production function of the pancreas (Figures 2A–C). Analysis of glucagon-producing cells demonstrated that *Acomys cahirinus* possessed 1.5-fold more glucagon-producing cells than *Mus musculus* (Figures 2A, D). Analysis of insulin-producing cells revealed that *Acomys cahirinus* possessed 2.5-fold more insulin-producing cells

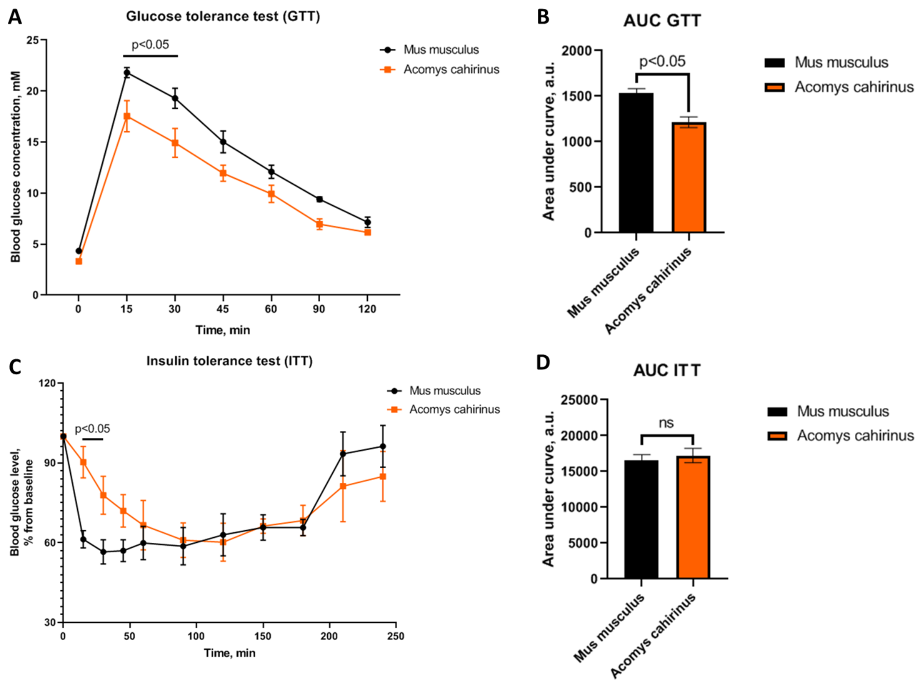

FIGURE 1
Acomys cahirinus has higher glucose tolerance and comparable insulin sensitivity relative to *Mus musculus*. **(A)** Blood glucose kinetics during the GTT; **(B)** AUC of the GTT; **(C)** blood glucose kinetics during the ITT; **(D)** AUC of the ITT. Data are presented as mean ± SEM (n = 10). Statistical significance was assessed using the Mann-Whitney rank sum U-test with a significance threshold of $p<0.05$.

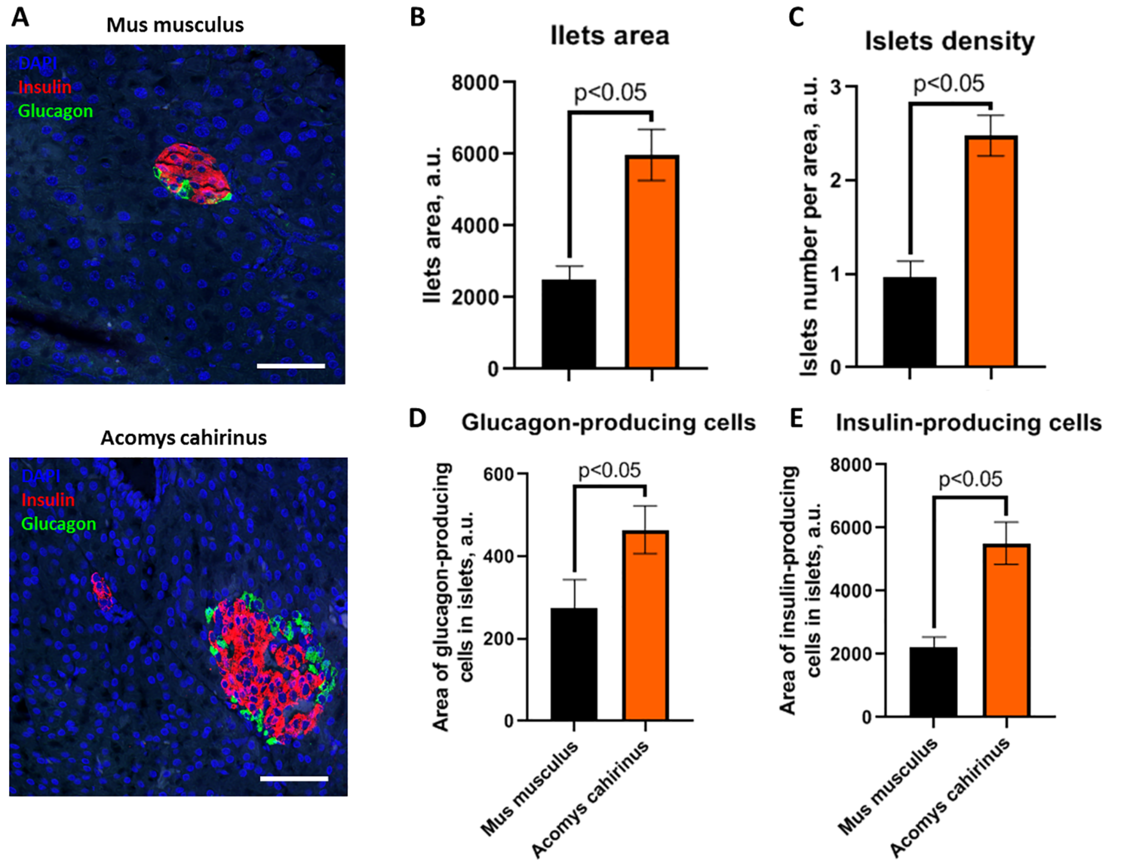

FIGURE 2
Acomys cahirinus exhibits higher islet area and density, with enhanced presence of insulin- and glucagon-producing cells compared with *Mus musculus*. **(A)** Representative images of Langerhans islet morphology, scale bar 100 um; **(B)** quantitative analysis of islet area in the pancreas; **(C)** quantitative analysis of islet density in the pancreas; **(D)** area of glucagon-producing cells within islets; **(E)** area of insulin-producing cells within islets. Data are presented as mean ± SEM (n = 10). Statistical significance was assessed using the Mann-Whitney rank sum U-test with a significance threshold of $p<0.05$.

than *Mus musculus* (Figures 2A, E). Thus, *Acomys cahirinus* has a predisposition to higher insulin production, which may contribute to its enhanced glucose clearance rate.

3.3 *Acomys cahirinus* exhibits suppressed gluconeogenesis based on classic substrates, but enhanced gluconeogenesis from protein-based substrates in the liver

The glucose clearance rate may be related not only to glucose uptake under insulin action but also to the regulation of glucose synthesis during hepatic gluconeogenesis. Based on this conclusion, we next analyzed the activity of glucose production on different substrates.

As a pilot experiment, we first analyzed hepatic glucose production ex vivo using liver explants. It has been shown that in conditions of empty media, *Acomys cahirinus* exhibited lower hepatic glucose production than *Mus musculus*, whereas pyruvate addition eliminated this difference (Figure 3A). Nevertheless, a significant difference in hepatic glucose production in the case of empty media prompted subsequent *in vivo* analysis of gluconeogenesis. It should be noted that these tolerance tests are not a clear assay for gluconeogenesis, because we analyzed glucose production, which can include not only gluconeogenesis but also potential glucose release or glucose uptake in any of the tissues. However, these assays can serve as an approximate *in vivo* assessment of gluconeogenic activity. We analyzed different intermediates of energetic biopolymers' ability to stimulate gluconeogenesis. As an intermediate of carbohydrate metabolism, pyruvate was selected, lipid metabolism - glycerol and protein metabolism - alanine. The PTT demonstrated that *Acomys cahirinus* produced less glucose in response to pyruvate administration than *Mus musculus* (Figures 3B, E), suggesting that *Acomys cahirinus* has less gluconeogenesis activity from carbohydrate-derived metabolites than *Mus musculus*. The GlyTT demonstrated lower glucose production in response to glycerol administration in *Acomys cahirinus* (Figures 3C, E). Because glycerol forms the backbone of triacylglycerols, this observation suggests reduced gluconeogenesis activity from lipid metabolites in *Acomys* cahirinus compared with *Mus musculus*. The ATT demonstrated that *Acomys cahirinus* produced more glucose in response to alanine administration (Figures 3D,E). This metabolic feature may represent an adaptation that enables *Acomys cahirinus* to use protein as a base for gluconeogenesis during prolonged starvation periods in arid ecosystems. Under basal laboratory conditions, suppressed gluconeogenesis from carbohydrates and lipids may also contribute to the lower

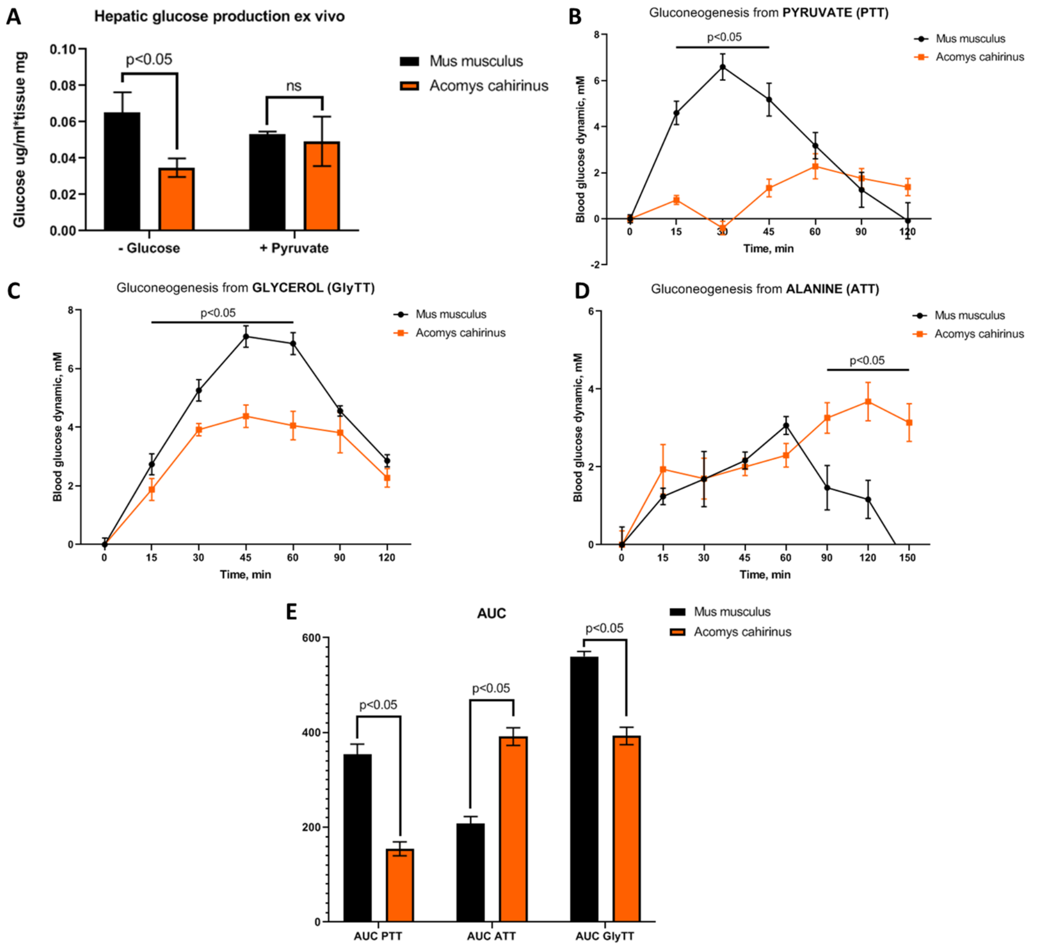

FIGURE 3
Acomys cahirinus exhibits suppressed gluconeogenesis on standard substrates, but enhanced gluconeogenesis on protein-based substrates. **(A)** Glucose production by liver explants; **(B)** glucose dynamic during the pyruvate tolerance test (PTT); **(C)** glucose dynamic during the glycerol tolerance test (GlyTT); **(D)** glucose dynamic during the alanine tolerance test (ATT); **(E)** comparison of AUC values during the PTT, ATT, and GlyTT. Data are presented as mean ± SEM, (n = 10). Statistical significance was assessed using the Mann-Whitney rank sum U-test with a significance threshold of p<0.05.

blood glucose levels and enhanced glucose tolerance observed in *Acomys cahirinus*.

3.4 *Acomys cahirinus* directs glucose flux in visceral fat depot with further metabolization to storage in lipids

The previous results demonstrated that *Acomys cahirinus* exhibits distinct mechanisms underlying its enhanced glucose clearance rate, including suppression of gluconeogenesis and enhancement of glucose uptake. However, the metabolic fate of the glucose uptake in *Acomys cahirinus* remained unclear. In this set of experiments, we analyzed the metabolic fate of uptaken glucose in the major fat depots of *Acomys cahirinus*.

First, we analyzed glucose uptake by different fat depots. Surprisingly, *Acomys cahirinus* exhibited significantly higher glucose uptake in epididymal fat, which is classic piece of visceral fat, whereas glucose uptake in subcutaneous fat was equal between *Acomys cahirinus* and *Mus musculus* (Figure 4A). To investigate the metabolic fate of the uptake glucose, we applied two strategies. As an indicator of glucose expenditure, we analyzed lactate concentration, one of the markers of glycolytic activity. As an indicator of glucose storage, we analyzed ^{14}C atoms incorporation into lipids. We found that lactate concentration in plasma was significantly lower in *Acomys cahirinus* after fasting, which may indicate active glycolysis in the case of *Mus musculus* (Figure 4B). In subcutaneous fat, lactate concentration was markedly lower in *Acomys cahirinus* than in *Mus musculus* in both fed and fasting states (Figure 4C). In visceral fat depot, lactate concentration was also significantly lower in *Acomys cahirinus* in the fed state (Figure 4D). Taken together, these findings suggest that *Acomys cahirinus* prefers to not use glucose in glycolysis, leading us to hypothesize that *Acomys cahirinus* may instead use uptaken glucose for lipid synthesis. Analysis of ^{14}C atoms incorporation from [^{14}C]-U-glucose demonstrated that [^{14}C]-labeled water-soluble metabolites did not differ in either subcutaneous or visceral fat depots in both animals (Figure 4E). The observation on [^{14}C]-labeled lipids revealed an interesting finding: [^{14}C]-labeled glucose was used for lipid synthesis in subcutaneous fat of *Mus musculus*, but in visceral fat of *Acomys cahirinus* (Figure 4F). Considering the increased glucose uptake observed in *the* visceral fat of *Acomys cahirinus*, these findings suggest that *Acomys cahirinus* prefers to use visceral

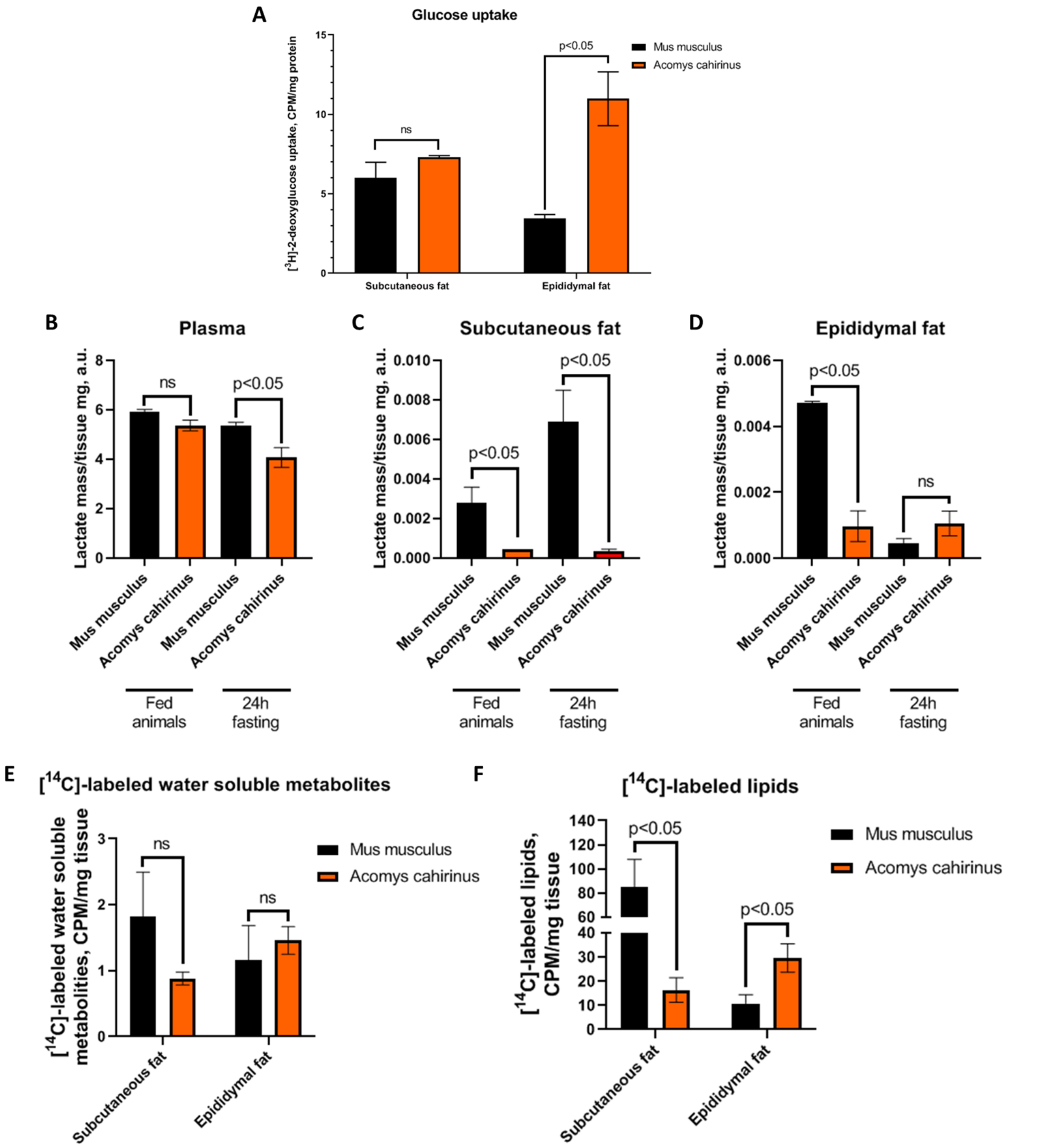

FIGURE 4
Acomys cahirinus predominantly takes up glucose into epididymal fat depots with further utilization in lipid synthesis. **(A)** Glucose uptake analysis in subcutaneous and visceral fat depots; **(B)** lactate concentration in plasma; **(C)** lactate concentration in subcutaneous fat; **(D)** lactate concentration in epididymal fat; **(E)** analysis of ^{14}C atoms incorporation from glucose in water-soluble metabolites; **(F)** analysis of ^{14}C atoms incorporation from glucose in lipids. Data are presented as mean ± SEM, (n = 5). Statistical significance was assessed using the Mann-Whitney rank sum U-test with a significance threshold of p<0.05.

fat as an energy depot, potentially contributing to the maintenance of enhanced glucose tolerance. This represents an unusual and original physiological adaptation contrary that contrasts with the classic theory of visceral fat in humans and classic model organisms. Based on previous results, our results indicate that *Acomys cahirinus* regulates glucose homeostasis through altered gluconeogenesis substrate specificity and lipid storage in visceral fat. Because the liver is the cross-talk organ that orchestrates both lipid and glucose metabolism, we analyzed selected aspects of glucose and lipid metabolism in the liver of *Acomys cahirinus*.

3.5 The liver of *Acomys cahirinus* exhibits a reversed glucose-lipid metabolism compared with *Mus musculus*, with high glycogen levels in the fed state, which transitions to high lipid levels in the fasting state

The liver is a central orchestrator of systemic metabolism of glucose and lipids. It serves as a glycogen depot that can be rapidly mobilized for energy expenditure processes. The liver also drives processes by lipid resynthesis and potentially their distribution to different tissues. We analyzed glycogen and lipid content in the liver of *Acomys cahirinus* histologically. For these experiments, animals were divided into fed and 24 h fasting groups. Each group (n = 10) was subdivided into a 24 h fasting group (n = 5) and a fed group (n = 5).

Analysis of glycogen content demonstrated that, in the fed state, the liver of *Acomys cahirinus* contained more glycogen than that of *Mus musculus* (Figures 5A, B). After 24 h of fasting, this pattern was reversed, with glycogen content becoming lower in the liver of *Acomys cahirinus* than in that of *Mus musculus* (Figures 5A, C). Thus, *Acomys cahirinus* uses glycogen for glucose homeostasis

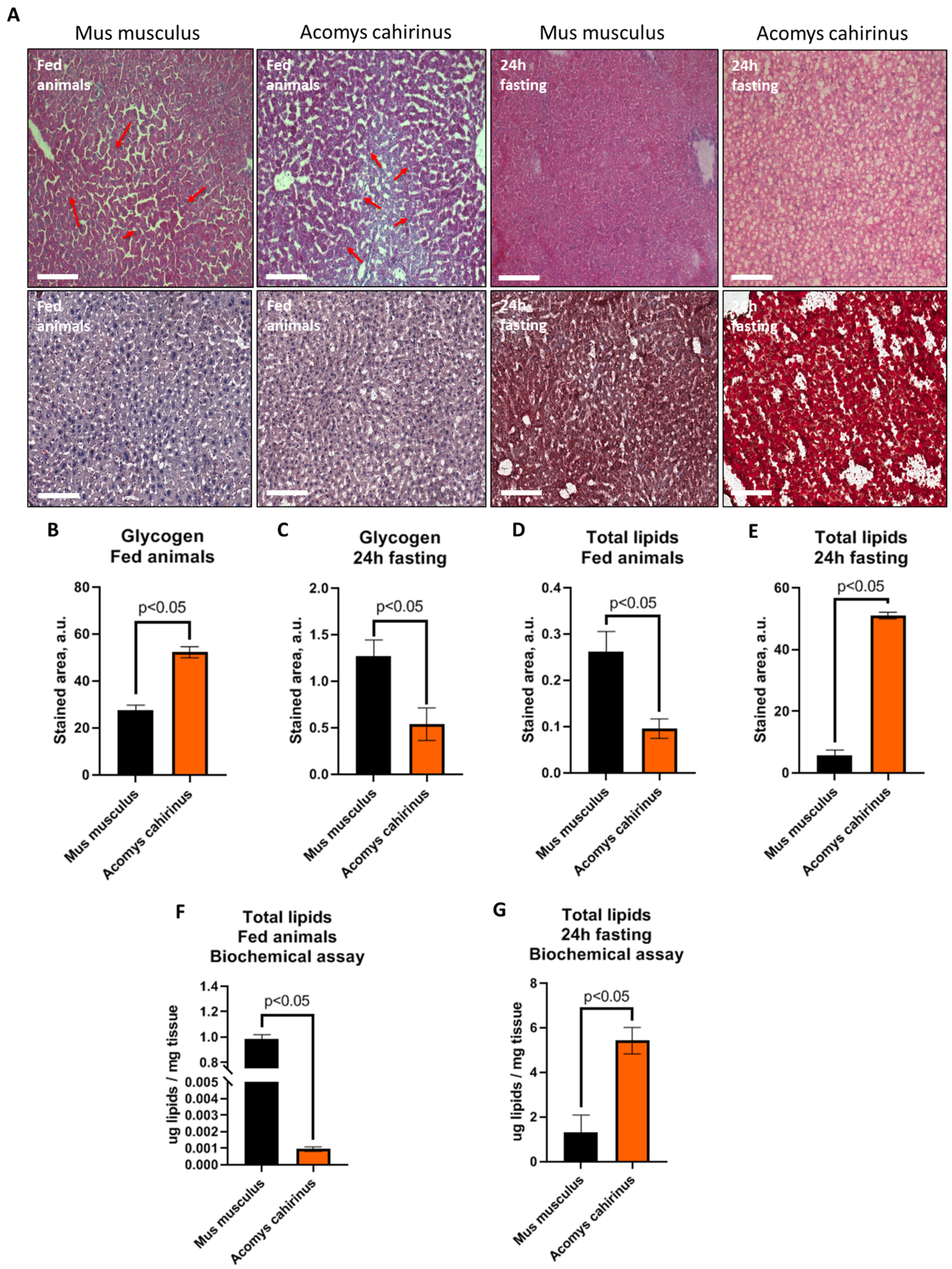

FIGURE 5
Acomys cahirinus exhibits reversed liver metabolism with high glycogen content in the fed state, with transition to high lipid content in the fasting state compared with *Mus musculus*. **(A)** Panel of representative images, upper panel - glycogen staining with Schiff's reagent, lower panel - lipid staining with Oil Red O; **(B)** quantitative analysis of glycogen content in the fed state; **(C)** quantitative analysis of glycogen content in the fasting state; **(D)** quantitative analysis of lipid content in the fed state; **(E)** quantitative analysis of lipid content in the fasting state; **(F)** biochemical lipid quantification in the fed state; **(G)** biochemical lipid quantification in the fasting state. Data are presented as mean ± SEM, (n = 5). Statistical significance was assessed using the Mann-Whitney rank sum U-test with a significance threshold of $p<0.05$.

maintenance during fasting periods. Analysis of lipid content demonstrated that, on the fed state, the liver of *Acomys cahirinus* contained lower lipid levels than that of *Mus musculus*, whereas, after 24 h of fasting, marked lipid accumulation was observed in the liver of *Acomys cahirinus* (Figures 5A, D, E). Such marked lipid accumulation in the liver of *Acomys cahirinus* was unexpected, and we analyzed lipid content in the liver using a biochemical phospho-sulfo-vanillin-based method. The results of the histological assay were fully confirmed (Figures 5F, G). In summary, *Acomys cahirinus* stores glycogen in the liver under fed conditions, whereas prolonged starvation periods, which are common in arid ecosystems, induce lipid accumulation in the liver.

4 Discussion

The present study investigated selected aspects of glucose and lipid metabolism in *Acomys cahirinus*, an animal from arid ecosystems that possesses high regenerative potential. Understanding the specific physiological adaptations of *Acomys cahirinus* may provide insights relevant to the development of new approaches in regenerative medicine and endocrinology. It should be noted that data regarding endocrinology and metabolic homeostasis amongin *Acomys cahirinus* are very limited. Therefore, in this discussion, we attempt to interpret the metabolic adaptations identified in this animal.

4.1 Enhanced glucose tolerance and pancreatic adaptations

First, we demonstrated enlarged glucose tolerance in *Acomys cahirinus* using systemic metabolic tests (Figure 1), which is consistent with our previous results showing lower fasting blood glucose levels in *Acomys cahirinus* (Boldyreva et al., 2025). Traditionally, *Acomys cahirinus* has been considered an animal that gradually develops impaired glucose tolerance with aging (Shafrir et al., 1972; Shafrir and Adler, 1983; Nesher and Cerasi, 1994; Shafrir et al., 2006). However, a serious limitation of these studies is that all of them were performed using the Jerusalem colony of spiny mice, which exhibit specific features compared with, for instance, the Geneva colony of spiny mice (Young, 1976; Haughton et al., 2016). Nevertheless, *Acomys cahirinus* from both colonies have been reported to display diabetic-like phenotype (Pictet et al., 1967; Gutzeit et al., 1974). In this context, the history of the discovery of the regenerative activity of *Acomys cahirinus* is particularly representative: its unique regenerative capacity was detected only in 2012 (Seifert et al., 2012; Maden and Varholick, 2020). Our results are in contrast with previous metabolic studies of *Acomys cahirinus*, and the potential mechanisms underlying this enhanced glucose tolerance required further investigation, which we have performed. We can hypothesize that enhanced glucose tolerance but lower insulin sensitivity during the early stage of the ITT can be explained by the ecology of *Acomys cahirinus*. *Acomys cahirinus* is a desert rodent, and the conditions of insufficient nutrient availability dictates the evolution of mechanisms that allow it to assimilate nutrients as rapidly as possible. In this context, enhanced glucose tolerance and independence of the uptake ability from insulin may suggest the presence of an insulin-independent mechanism that promotes the rapid uptake of nutrients from the bloodstream of *Acomys cahirinus*.

Enhanced glucose tolerance and consequently high glucose clearance rate are traditionally associated with pancreatic activity. We demonstrated that the pancreas of *Acomys cahirinus* contains a greater number of Langerhans islets, and that insulin- and glucagon-producing cell numbers were significantly higher in the case of *Acomys cahirinus* (Figure 2). It should be emphasized that our findings postulate only a difference in the morphological determinants of insulin and glucagon production, not in the circulating level of respective hormones. These results are consistent with previous results, which demonstrated hyperplastic pancreas remodeling in the case of *Acomys cahirinus* (Gonet et al., 1966; Pictet et al., 1967). These studies indicated the development of hyperinsulinemia, which was associated with hyperglycemia. However, despite pancreatic hyperplastic remodeling, in our study, we observed enhanced glucose tolerance in *Acomys cahirinus*, which could be a consequence of the enhanced peripheral tissue insulin sensitivity observed in our study in comparison with earlier reports.

4.2 Hepatic gluconeogenesis: a shift in substrate preference

The pancreas is not the only organ involved in glucose homeostasis maintenance. The liver is a central orchestrator of systemic glucose metabolism, which also contributes to glucose homeostasis through the regulation of the gluconeogenesis rate. Gluconeogenesis can be arranged by different substrates: pyruvate from carbohydrates, glycerol from lipids, and amino acid carbon skeletons from proteins. Lipid-based gluconeogenesis is one of the characteristics of several Arctic animals, for instance, ground squirrels (*Citellus undulatus*) and northern elephant seals (*Mirounga angustirostris*) (Burlington and Klain, 1967; Galster and Morrison, 1975; Kaleta et al., 2012). Nevertheless, protein-based gluconeogenesis is also present among ground squirrels, animals with limited access to food (Galster and Morrison, 1975). Classic animals of arid ecosystems, such as antelopes (*Oryx leucoryx*) and camels (*Camelus dromedarius*), prefer to use active gluconeogenesis from carbohydrates during prolonged fasting periods (Emmanuel, 1981; Ostrowski et al., 2006). Protein-based gluconeogenesis has been poorly characterized in desert animals and is more commonly associated with obligate carnivores and ruminants (Young, 1977; Nafikov and Beitz, 2007; Verbrugghe and Hesta, 2017). *Acomys cahirinus* exhibits a particularly interesting gluconeogenesis substrate profile. Spiny mice demonstrated reduced gluconeogenesis from classic desert-animal substrates and suppressed fat-based gluconeogenesis, together with enhanced protein-based gluconeogenesis (Figure 3). It is possible that, under basal conditions, protein gluconeogenesis can be suppressed and low total gluconeogenic activity in the liver from classic substrates contributes to the maintenance of high glucose tolerance in *Acomys cahirinus*. It should be noted that prolonged starvation induces some mammals to switch to protein-based gluconeogenesis (Galster and Morrison, 1975).

4.3 Visceral adipose tissue as a primary glucose sink and energy depot

The glucose uptake profile can show the relative metabolic activity of different tissues at the whole-body level. In our previous study, we observed enhanced development of adipose tissue in *Acomys cahirinus* (Boldyreva et al., 2025). In the present study, we analyzed glucose uptake distribution in subcutaneous and visceral fat depots. Unlike *Mus musculus*, which predominantly stores fat in subcutaneous depots, *Acomys cahirinus* exhibited a marked shift toward visceral fat deposition (Figure 4A). Although visceral adiposity is associated with pathological consequences in human clinical physiology, in *Acomys cahirinus* this trait may represent an adaptive strategy (Ehrhardt et al., 2005; Sakers et al., 2022). Visceral fat is characterized by richer vascularization and innervation, together with greater sensitivity to catecholamines than subcutaneous depots, enabling the rapid release of free fatty acids into the bloodstream (Wajchenberg, 2000; Frayn and Karpe, 2014). This metabolic organization allows for the rapid mobilization of energy under conditions of unpredictable food availability. Moreover, visceral fat deposition plays a major role in thermoregulation of desert animals by minimizing subcutaneous fat depot and maximizing heat dissipation (Schmidt-Nielsen, 1959; Stenvinkel et al., 2026).

The metabolic fate of the uptaken glucose was also highly interesting. *Acomys cahirinus* preferentially avoided metabolizing uptaken glucose in glycolysis, and tissue lactate levels were markedly reduced compared with *Mus musculus* (Figures 4B–D). It is also an adaptation of hibernating and desert animals that face low food accessibility during hibernation or arid ecosystems. In this case, the glycolysis rate significantly decreased with the switch to fat-based metabolism (Storey, 1997; Carey et al., 2003; Johnson et al., 2025). The [^{14}C]-tracing assay demonstrated a classic pattern of desert animals. *Mus musculus* accumulated lipids in subcutaneous fat, but not in visceral fat depot, whereas *Acomys cahirinus* did not accumulate lipids in subcutaneous fat and instead metabolized glucose to lipids within visceral fat depot, which is beneficial for maximizing heat release and fat storage for potential metabolic water supplementation (Figures 4E,F) (Johnson et al., 2013; Olsen et al., 2021).

4.4 Liver metabolic flexibility: from glycogen storage to lipid accumulation

The liver is a central orchestrator of systemic metabolism, and the hepatic metabolic adaptations of *Acomys cahirinus* are particularly relevant to understanding enhanced glucose homeostasis. We demonstrated that, in the fed state, the liver of *Acomys cahirinus* contains high glycogen content, which expands during fasting and transforms into lipid form, whereas Mus musculus exhibited the opposite mechanism (Figure 5). High glycogen content in the liver is a well-known adaptation in desert animal, including camels (*Camelus dromedarius*) and ground squirrels (*Citellus undulatus*) (Bintz et al., 1979; Candlish, 1981; Emmanuel, 1981). In *Acomys cahirinus*, we observed the transformation of high glycogen to high-lipid content during starvation, and previous studies have confirmed a predisposition of the liver of *Acomys cahirinus* to hyperlipidemia (Shafrir and Adler, 1983). On the other hand, starvation can stimulate lipid accumulation, which is observed in humans and classic diabetic models (van Ginneken, 2008). During analysis of liver physiology of many desert animals, it has been reported that all of them have an enlarged rate of hepatic steatosis formation and ectopic fat disposition in the liver, including camels (*Camelus dromedarius*) and rodents (*Psammomys obesus*, *Gerbil gerbillus*) (Lalla and Drommer, 1997; Spolding et al., 2014; Semiane et al., 2017; Musina et al., 2025). The balance between glucose and lipid accumulation, which animals use for storage of metabolic water and as an adaptation to low-food accessibility in the vivarium conditions, usually transforms from adaptation to pathology, and if in wild conditions, such lipid storage mechanisms during starvation periods protect *Acomys cahirinus* from hypoglycemia and dehydration, in laboratory conditions, they can predispose the animals o disease.

5 Conclusion

In summary, the main findings of the present study are: (i) enhanced glucose tolerance in *Acomys cahirinus*, associated with (ii) pancreas hyperplasia, (iii) suppressed hepatic gluconeogenesis on standard substrates, and (iv) predominant visceral fat glucose uptake and lipogenesis. All these mechanisms contribute to the evolutionary success of *Acomys cahirinus* in the harsh conditions of arid ecosystems. *Acomys cahirinus* is also widely recognized for its high regenerative capacity, and the features presented in this study may be relevant in the context of regeneration. At present, studies on regeneration and metabolism remain limited. Evidence presents two opposite perspectives. The first is from the analysis of cardiac cells, which have suggested that glycolytic metabolism is necessary in the hearts of *Acomys cahirinus* for post-infarction regeneration (Kuppa et al., 2024). The second is from the analysis of fibroblasts, which have reported on active mitochondrial and lipid metabolism related to fibroblast regeneration (Aryee et al., 2025). Our results are closer to the results from the fibroblast study, but direct comparisons between these studies should be made cautiously because they involve distinct tissues.

Acomys cahirinus is a desert rodent that requires specific adaptations to arid ecosystems. *Acomys cahirinus* has enhanced glucose tolerance, which is realized through a hyperplastic pancreas, changes in substrate predisposition during hepatic gluconeogenesis, and predominant glucose uptake in the visceral fat depot followed by lipogenesis. Some of these mechanisms may be used as a potential strategy for human metabolic disease treatment and warrant further investigation.

6 Limitations

As a classic study with comparisons between different biological species, our study has some limitations that should be acknowledged. First, the animals used in this study differed in body weight;

however, this can be explained by the fact that animals of the same post-puberty age were used. Second, the diet of *Acomys cahirinus* was supplemented with dried *Hermetia illucens* imagoes (three per week per animal). This supplementation can be explained because *Acomys cahirinus* is a semi-insectivorous species; and dried *Hermetia illucens* was added to family cages (<5% of the diet) to support reproduction, regeneration, and calcium-phosphorus balance in pregnant and lactating females. *Mus musculus* did not receive this supplementation because standard pellet diets are considered fully balanced.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The animal study was approved by Ethical Board of the Chazov National Medical Research Center of Cardiology. The study was conducted in accordance with the local legislation and institutional requirements.

Author contributions

MB: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. RM: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. MA: Investigation, Visualization, Writing – review & editing. SM: Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing. NA: Formal analysis, Methodology, Writing – review & editing. ER: Investigation, Project administration, Resources, Writing – review & editing. YP: Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. IS: Conceptualization, Data curation, Investigation, Supervision, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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